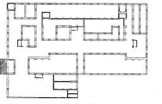


PROJECT TITLE : M15X Ph2 Project

[illegible]

ITEM NO.	W14-SU-4721
SERVICE	REUSE DRAIN TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	W x L x H (m) 5.0 x 2.5 x 2.5 (EH 2.0)
CAPA	m3 31.25 (EH 25.0)
MAT'L	R.C + EPOXY

대외비



KEY PLAN (1F)

- NOTE
1. REUSE 탱크 내부 차폐 배기.
 2. INTERLOCK NO. 1-7028
(W14-SPP-4721 A/B AUTO START)
- REMOTE STATUSMAN W14-SU-4721 LEVEL
(HS-W7019) HIGHMAN W14-SPP-4721 A/B START
(HS-W7019) HIGH HIGHMAN W14-SPP-4721 A, B
BOTH PUMPS START.
INTERLOCK NO. 1-7028
(W14-SPP-4721 A/B AUTO STOP)
- REMOTE STATUSMAN W14-SU-4721 LEVEL
(LS-W7019) LOW LOWMAN W14-SPP-4721 A, B
BOTH PUMPS STOP.
 3. 콘크리트 CLOSE 리프 배관에는 LOTO 설치.
 4. W14-SPP-4721 A/B 탱크 선별 콘센트.
 5. CHECKED PLATE COVER.
 6. SUMP PUMP의 작동배관은 상부호 연결이다(배관부).
 7. PUMP의 SUMP DRAW-OFF PIT 내부 설치됨.
 8. SUMP INTERNAL LADDER 규격함.

D-636

636

641

OD110-D-HP1-N-7038

HWL : B + 2.2
HML : B + 2.0
LWL : B + 0.2
LLWL : B + 0.0

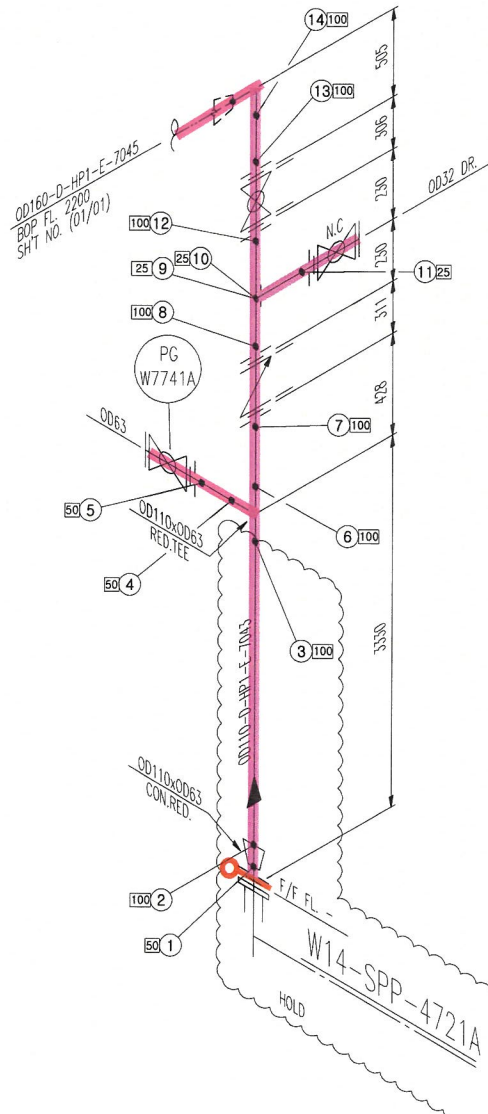
ITEM NO.	W14-SPP-4721 A/B
SERVICE	REUSE DRAIN TANK PUMP
Q'TY	1 + 1
TYPE	SUBMERSIBLE
CAPA	m3/hr ea x mH 45 x 30 (0.29MPa)
MAT'L	SSC13/SSC13
POWER	BY VENDOR

W14-SU-4721
NOTE 1

W14-SPP-4721 A W14-SPP-4721 B
NOTE 4,7

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 DRAWN BY	이민혁	
검 CHECKED BY	한진희	
승 APPROVED BY	박동영	
제 작 사	SK 에코플랜트	
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project	
도 면 명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM REUSE DRAIN TANK & PUMP	
축 척 SCALE	NONE	
날 짜 DATE	2025. 08. 14	
PROJECT NO.	223300	REV.
도면번호 SHEET NO.	CJ-W15W14-PR-A20-000320	1A

3



<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는

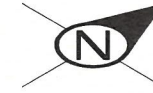
2. ELEVATION, COUPLING, HANGAR, FIELD ADJUSTING

3. CHECK VALVE, VC/VCT/VCT2/VC3/VC4/VC5/VC6/VC7

4. SPECIAL BOLT & NUT : BL/BL1

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RCS/RCS1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1
 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : BF/BF1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB0/VB1/VB10/VB2/VB20 * BUTTERFLY VA. : VF/VF4
 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VD0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXH * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
 * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R1FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-010016-A	REF. P & ID	CJ-M15W14-PR-A20-000320	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60														
			OPERATING	4	AMB														



SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
HP1			
P	OD32	0.1M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD63	0.1M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD110	4.7M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
E90	OD110	1EA	ELBOW 90 DEG, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
TR	OD110xOD63	1EA	TEE RED, HDPE PE100, PE, BUTT FUSION X SOCKET FUSION, PN16/SDR11
RC	OD110xOD63	1EA	REDUCER CON, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
SD	OD110xOD32	1EA	SADDLE, HDPE PE100, PE, BRANCH-SOCKET FUSION, PN16/SDR11
VB	OD32	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER
VB	OD63	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER
VB	OD110	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER
VC	OD110	1EA	CHECK VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP, BC, SWING
F	OD32	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
F	OD63	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
F	OD110	4EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-ST304 / STUB END-HDPE PE100
FB	OD32	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD63	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD110	4EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B	OD32	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	OD63	3SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	OD110	4SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
PG	OD63	1EA	PRESSURE GAUGE, KS 10K, FF
LOTO	OD32	1EA	LOTO
IN	25xOD110	4.7M	관 보편(200mm)
FB	OD63	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-ST304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)

CLIENT

SK hynix

1

CONTRACTOR

SK ecoplant

PROJECT NAME

M15X Ph4 Project

COMMODITY

D

JOB NO.

22330D

LINE NO.

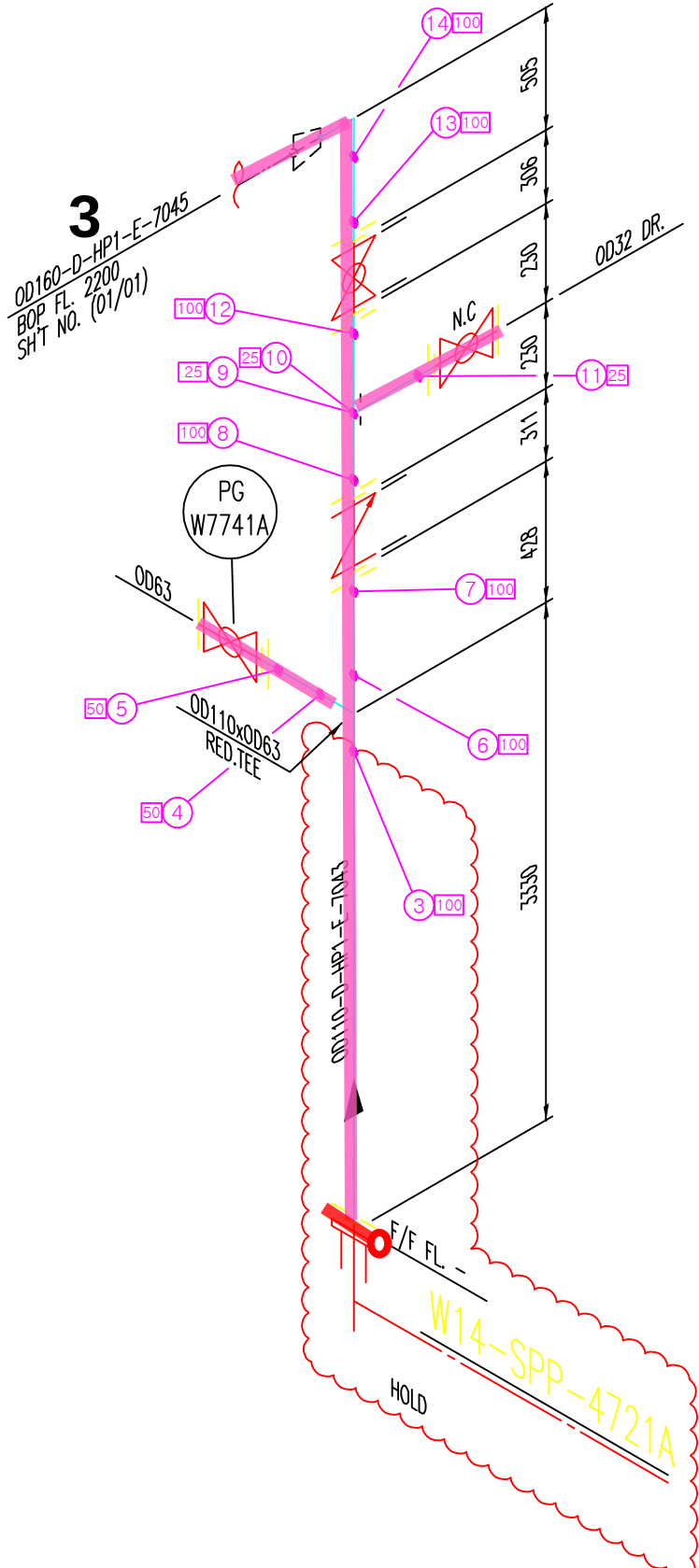
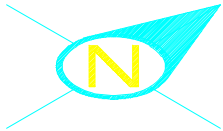
OD110-D-HP1-E-7043

SHT NO.

01 / 01

REV.

0



<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는

* PIPE ELEVATION COUPLING 및 그 이하의 COUPLING : HF/HF1

2. (*) FIELD ADJUST

* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7

* BUTTERFLY VA. : VF/VF4

* STEAM TRAP : ST

* SPRAY NOZZLE : SPN

* EMERGENCY SHOWER : SSH

* SPECIAL BOLT & NUT : BL/BL1

* PRESSURE GAUGE : PG

* SHILED : SH

* SIGHT GLASS : SG

* LOTO : LOTO

* AIR TRAP : AT

* ISOLATION KIT : IK

* STRAINER(BUCKET-TYPE) : BSTR

* STRAINER(T-TYPE) : TSTR

* STRAINER(Y-TYPE) : YSTR

* GLOBE VA. : VL/VL1/VL2

* DIAPHRAGM VA. : VD/VDLO

* DAMPER : VP

* QUICK COUPLING : SQC

* FLEXIBLE JOINT(STEEL) : FLXB

* FLEXIBLE JOINT(RUBBER) : FLXR

* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL

* FLEXIBLE HOSE : FLXH

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* GATE VA. : VG/VG1/VG2

* BLIND FLANGE : FB/FB1

* BOLT & NUT : B/B1

* GASKET : G/G1/G2

* FLANGE : F/F1

* SADDLE(HDPE) : SD

* SOCKET(HDPE/PVC) : SC

* STUB-IN : STU

* RED.TEE : TR

* TEE : T

* 45° ELBOW : E45

* 90° ELBOW : E90

* PIPE : P/P1

* CON.RED : RC

* ECC.RED : RE

* CON.SWAGE : RCS/RCS1

* ECC.SWAGE : RES/RES1

* NIPPLE : NP/NP1/NP2

* CAP : CAP/CAP1

* REINFORCING PAD : RP

* STUB-IN : STU

* SOCKET(HDPE/PVC) : SC

* SADDLE(HDPE) : SD

* FLANGE : F/F1

* GASKET : G/G1/G2

* BOLT & NUT : B/B1

* BLIND FLANGE : FB/FB1

* GATE VA. : VG/VG1/VG2

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* BUTTERFLY VA. : VF/VF4

* STEAM TRAP : ST

* SPRAY NOZZLE : SPN

* EMERGENCY SHOWER : SSH

* SPECIAL BOLT & NUT : BL/BL1

* PRESSURE GAUGE : PG

* SHILED : SH

* SIGHT GLASS : SG

* LOTO : LOTO

* AIR TRAP : AT

* ISOLATION KIT : IK

* STRAINER(BUCKET-TYPE) : BSTR

* STRAINER(T-TYPE) : TSTR

* STRAINER(Y-TYPE) : YSTR

* GLOBE VA. : VL/VL1/VL2

* DIAPHRAGM VA. : VD/VDLO

* DAMPER : VP

* QUICK COUPLING : SQC

* FLEXIBLE JOINT(STEEL) : FLXB

* FLEXIBLE JOINT(RUBBER) : FLXR

* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL

* FLEXIBLE HOSE : FLXH

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* GATE VA. : VG/VG1/VG2

* BLIND FLANGE : FB/FB1

* BOLT & NUT : B/B1

* GASKET : G/G1/G2

* FLANGE : F/F1

* SADDLE(HDPE) : SD

* SOCKET(HDPE/PVC) : SC

* STUB-IN : STU

* RED.TEE : TR

* TEE : T

* 45° ELBOW : E45

* 90° ELBOW : E90

* PIPE : P/P1

* CON.RED : RC

* ECC.RED : RE

* CON.SWAGE : RCS/RCS1

* ECC.SWAGE : RES/RES1

* NIPPLE : NP/NP1/NP2

* CAP : CAP/CAP1

* REINFORCING PAD : RP

* STUB-IN : STU

* SOCKET(HDPE/PVC) : SC

* SADDLE(HDPE) : SD

* FLANGE : F/F1

* GASKET : G/G1/G2

* BOLT & NUT : B/B1

* BLIND FLANGE : FB/FB1

* GATE VA. : VG/VG1/VG2

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* BUTTERFLY VA. : VF/VF4

* STEAM TRAP : ST

* SPRAY NOZZLE : SPN

* EMERGENCY SHOWER : SSH

* SPECIAL BOLT & NUT : BL/BL1

* PRESSURE GAUGE : PG

* SHILED : SH

* SIGHT GLASS : SG

* LOTO : LOTO

* AIR TRAP : AT

* ISOLATION KIT : IK

* STRAINER(BUCKET-TYPE) : BSTR

* STRAINER(T-TYPE) : TSTR

* STRAINER(Y-TYPE) : YSTR

* GLOBE VA. : VL/VL1/VL2

* DIAPHRAGM VA. : VD/VDLO

* DAMPER : VP

* QUICK COUPLING : SQC

* FLEXIBLE JOINT(STEEL) : FLXB

* FLEXIBLE JOINT(RUBBER) : FLXR

* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL

* FLEXIBLE HOSE : FLXH

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* GATE VA. : VG/VG1/VG2

* BLIND FLANGE : FB/FB1

* BOLT & NUT : B/B1

* GASKET : G/G1/G2

* FLANGE : F/F1

* SADDLE(HDPE) : SD

* SOCKET(HDPE/PVC) : SC

* STUB-IN : STU

* RED.TEE : TR

* TEE : T

* 45° ELBOW : E45

* 90° ELBOW : E90

* PIPE : P/P1

* CON.RED : RC

* ECC.RED : RE

* CON.SWAGE : RCS/RCS1

* ECC.SWAGE : RES/RES1

* NIPPLE : NP/NP1/NP2

* CAP : CAP/CAP1

* REINFORCING PAD : RP

* STUB-IN : STU

* SOCKET(HDPE/PVC) : SC

* SADDLE(HDPE) : SD

* FLANGE : F/F1

* GASKET : G/G1/G2

* BOLT & NUT : B/B1

* BLIND FLANGE : FB/FB1

* GATE VA. : VG/VG1/VG2

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* BUTTERFLY VA. : VF/VF4

* STEAM TRAP : ST

* SPRAY NOZZLE : SPN

* EMERGENCY SHOWER : SSH

* SPECIAL BOLT & NUT : BL/BL1

* PRESSURE GAUGE : PG

* SHILED : SH

* SIGHT GLASS : SG

* LOTO : LOTO

* AIR TRAP : AT

* ISOLATION KIT : IK

* STRAINER(BUCKET-TYPE) : BSTR

* STRAINER(T-TYPE) : TSTR

* STRAINER(Y-TYPE) : YSTR

* GLOBE VA. : VL/VL1/VL2

* DIAPHRAGM VA. : VD/VDLO

* DAMPER : VP

* QUICK COUPLING : SQC

* FLEXIBLE JOINT(STEEL) : FLXB

* FLEXIBLE JOINT(RUBBER) : FLXR

* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL

* FLEXIBLE HOSE : FLXH

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* GATE VA. : VG/VG1/VG2

* BLIND FLANGE : FB/FB1

* BOLT & NUT : B/B1

* GASKET : G/G1/G2

* FLANGE : F/F1

* SADDLE(HDPE) : SD

* SOCKET(HDPE/PVC) : SC

* STUB-IN : STU

* RED.TEE : TR

* TEE : T

* 45° ELBOW : E45

* 90° ELBOW : E90

* PIPE : P/P1

* CON.RED : RC

* ECC.RED : RE

* CON.SWAGE : RCS/RCS1

* ECC.SWAGE : RES/RES1

* NIPPLE : NP/NP1/NP2

* CAP : CAP/CAP1

* REINFORCING PAD : RP

* STUB-IN : STU

* SOCKET(HDPE/PVC) : SC

* SADDLE(HDPE) : SD

* FLANGE : F/F1

* GASKET : G/G1/G2

* BOLT & NUT : B/B1

* BLIND FLANGE : FB/FB1

* GATE VA. : VG/VG1/VG2

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* BUTTERFLY VA. : VF/VF4

* STEAM TRAP : ST

* SPRAY NOZZLE : SPN

* EMERGENCY SHOWER : SSH

* SPECIAL BOLT & NUT : BL/BL1

* PRESSURE GAUGE : PG

* SHILED : SH

* SIGHT GLASS : SG

* LOTO : LOTO

* AIR TRAP : AT

* ISOLATION KIT : IK

* STRAINER(BUCKET-TYPE) : BSTR

* STRAINER(T-TYPE) : TSTR

* STRAINER(Y-TYPE) : YSTR

* GLOBE VA. : VL/VL1/VL2

* DIAPHRAGM VA. : VD/VDLO

* DAMPER : VP

* QUICK COUPLING : SQC

* FLEXIBLE JOINT(STEEL) : FLXB

* FLEXIBLE JOINT(RUBBER) : FLXR

* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL

* FLEXIBLE HOSE : FLXH

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* GATE VA. : VG/VG1/VG2

* BLIND FLANGE : FB/FB1

* BOLT & NUT : B/B1

* GASKET : G/G1/G2

* FLANGE : F/F1

* SADDLE(HDPE) : SD

* SOCKET(HDPE/PVC) : SC

* STUB-IN : STU

* RED.TEE : TR

* TEE : T

* 45° ELBOW : E45

* 90° ELBOW : E90

* PIPE : P/P1

* CON.RED : RC

* ECC.RED : RE

* CON.SWAGE : RCS/RCS1

* ECC.SWAGE : RES/RES1

* NIPPLE : NP/NP1/NP2

* CAP : CAP/CAP1

* REINFORCING PAD : RP

* STUB-IN : STU

* SOCKET(HDPE/PVC) : SC

* SADDLE(HDPE) : SD

* FLANGE : F/F1

* GASKET : G/G1/G2

* BOLT & NUT : B/B1

* BLIND FLANGE : FB/FB1

* GATE VA. : VG/VG1/VG2

* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO

* BUTTERFLY VA. : VF/VF4

* STEAM TRAP : ST

* SPRAY NOZZLE : SPN

* EMERGENCY SHOWER : SSH

* SPECIAL BOLT & NUT : BL/BL1

* PRESSURE GAUGE : PG

* SHILED : SH

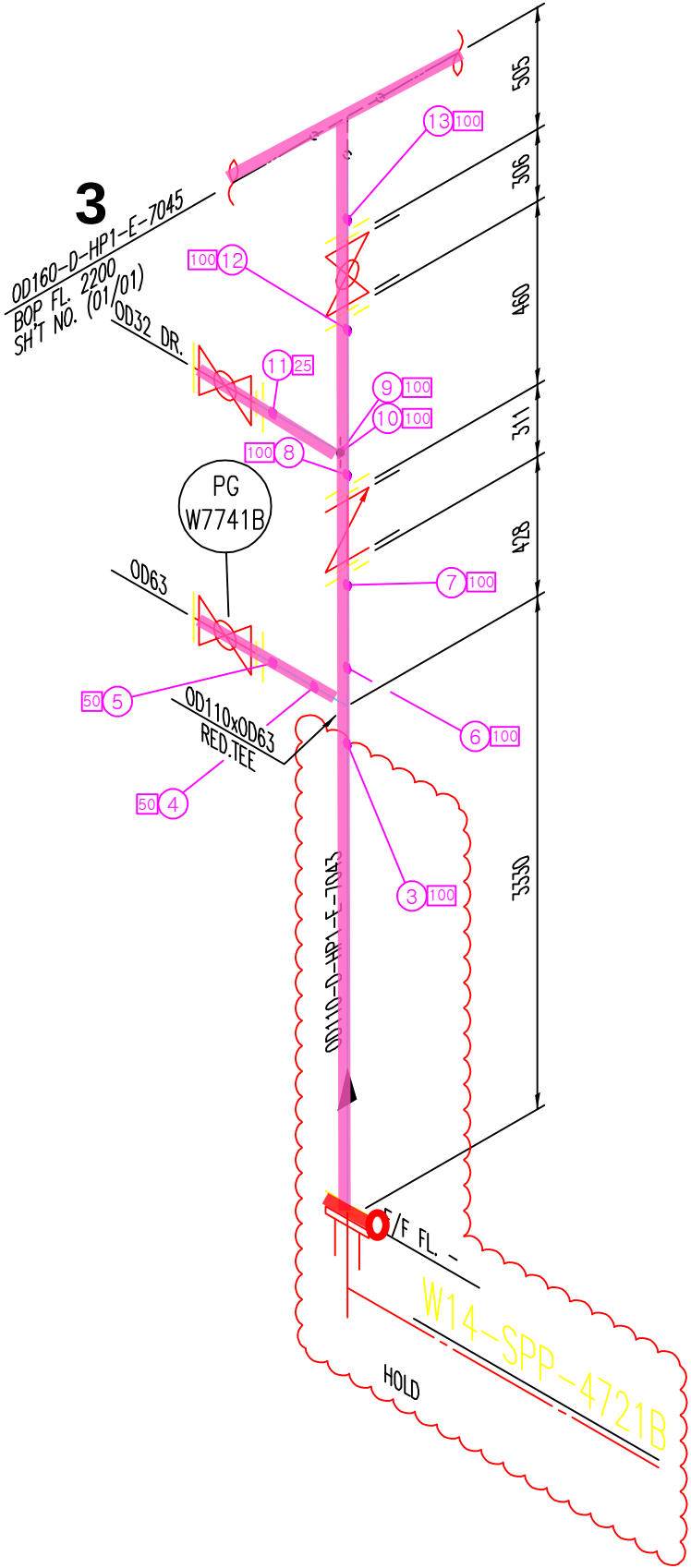
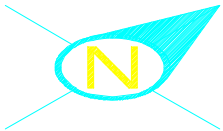
* SIGHT GLASS : SG

* LOTO : LOTO

* AIR TRAP : AT

[illegible]

CLIENT			 2		
			PROJECT NAME		
			M15X Ph4 Project		
COMMODITY			JOB NO.		
D			22330D		
LINE NO.	0D110-D-HP1-E-7044		SH'T NO.	REV.	
			01 / 01	0	



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
HP1			
P	OD32	0.1M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD63	0.1M	PIPE, HDPE PE100, PE, SOCKET FUSION, PN16/SDR11
P	OD110	4.7M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
TR	OD110xOD63	1EA	TEE RED, HDPE PE100, PE, BUTT FUSION X SOCKET FUSION, PN16/SDR11
SD	OD110xOD32	1EA	SADDLE, HDPE PE100, PE, BRANCH-SOCKET FUSION, PN16/SDR11
VB	OD32	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER
VB	OD63	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER
VB	OD110	1EA	BALL VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP / SEAT-PTFE, FB, LEVER
VC	OD110	1EA	CHECK VALVE, KS 10K, FF FLG'D, BODY-PP / TRIM-PP, BC, SWING
F	OD32	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STSS304 / STUB END-HDPE PE100
F	OD63	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STSS304 / STUB END-HDPE PE100
F	OD110	4EA	FLG LAP JOINT / STUB END SET, KS 10K, FF, LAP JOINT-STSS304 / STUB END-HDPE PE100
FB	OD32	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STSS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)
G	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD63	3EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD110	4EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G2	OD32	1EA	GASKET, KS 10K, FF, PTFE+EPDM, BLIND, 3.0MM TH'K
B	OD32	2SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	OD63	3SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
B	OD110	4SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
PG	OD63	1EA	PRESSURE GAUGE, KS 10K, FF
LOTO	OD32	1EA	LOTO
IN	25tXOD110	4.7M	관 보온재(고무발포)
FB	OD63	1EA	FLG BLIND & LINER SET, KS 10K, FF, BLIND-STSS304 / LINDER-HDPE PE100 (MIN 5.0MM TH'K)

CLIENT

SK hynix

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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									
	THICK.	-	DESIGN	6.4	60	UNIT	R1FA									
			OPERATING	4	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-010016-A									
						REF. P. & ID	CJ-M15W14-PR-A20-000320									
						REV. NO.	25.01.24									
						DATE	APPROVED FOR CONSTRUCTION									
						DESCRIPTION	황래중									
						BY	안남후									
						CHK	박동영									
						APP'D										

CONTRACTOR

SK ecoplant

PROJECT NAME

M15X Ph4 Project

COMMODITY

D

JOB NO.

22330D

LINE NO.

OD110-D-HP1-E-7044

SH'T NO.

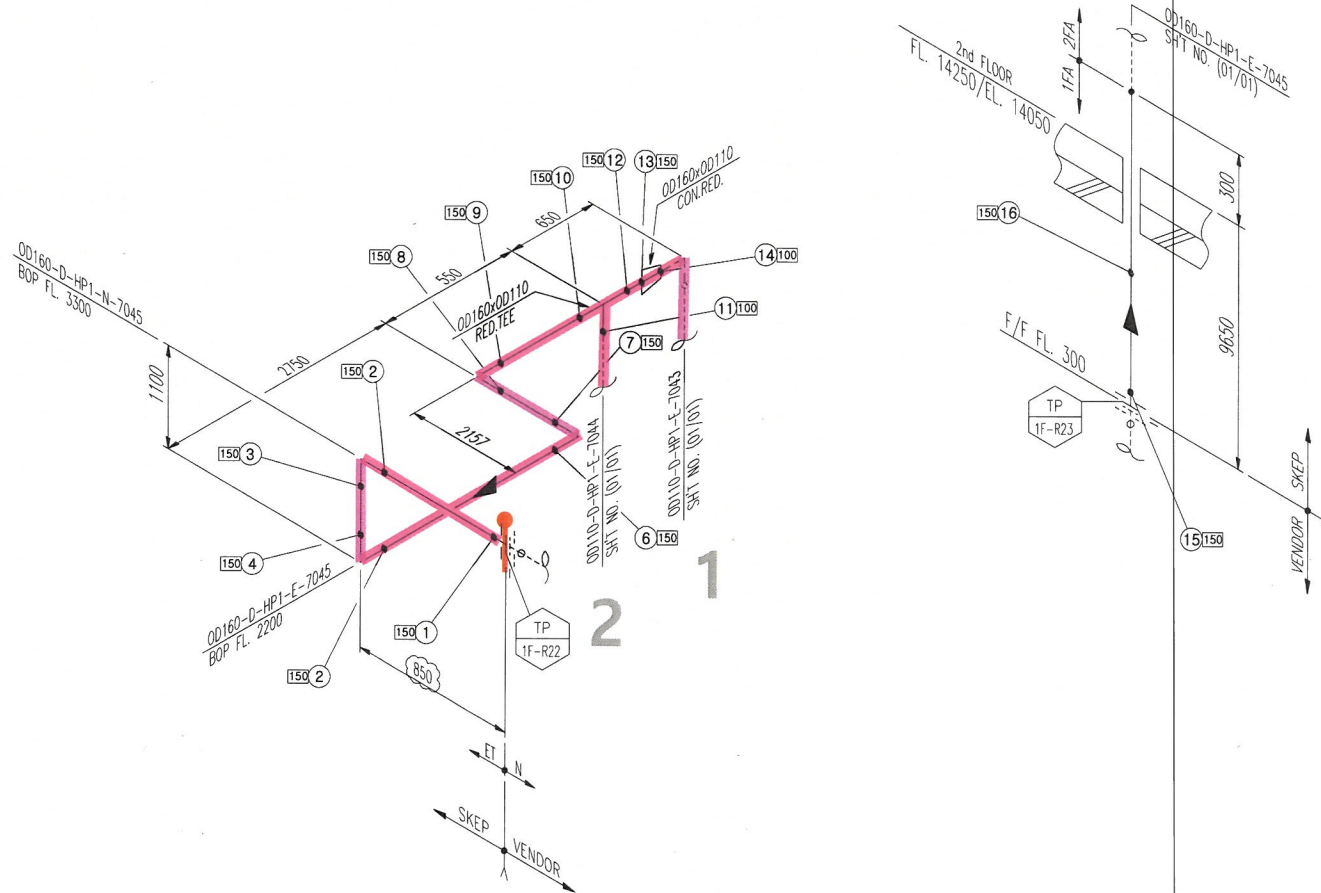
01 / 01

REV.

0



ITEM	SIZE	Q'TY	DESCRIPTION
P	OD160	19.2M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
E90	OD160	4EA	ELBOW 90 DEG, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
TR	OD160xOD110	1EA	TEE RED, HDPE PE100, PE, BUTT FUSION X SOCKET FUSION, PN16/SDR11
RC	OD160xOD110	1EA	REDUCER CON, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
IN	251xOD160	7.9M	관 보온재(고무발포)
F	OD160	2EA	FLG LAP JOINT / STUB END SET, KS 10K, FT, LAP JOINT-ST5304 / STUB END-HDPE PE100



<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT기 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CONV. RED : RC * ECC RED : RE * CON. SWAGE : RCS/RCS1 * ECC SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SOC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14	UNIT	R1FA	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60				25.03.31	REVISED AS MARKED	황래중	안남후	박동영	
			OPERATING	4	AMB				25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	
						REF. PLAN DWG. CJ-M15W14-PI-A30-010017-B/C								
						REF. P & ID CJ-M15W14-PR-A20-000320,000501								

CLIENT	SK hynix	3
CONTRACTOR	SK ecoplant	PROJECT NAME M15X Ph4 Project
COMMODITY	D	JOB NO. 22330D
LINE NO.	OD160-D-HP1-E-7045	SH'T NO. 01 / 01
		REV. 1

PUNCH LIST

Project No : 22330D
Project Title : M15X Ph3 Project

Package No :
Date :

S/N	Drawing No	Description		Category A B C	Punch Clear Status			
					작성자		확인자	
					Name	Sign	Name	Sign
1								
2								
3								
4								
5								
6								
7								
8								
9								
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14								
15								

Remarks

A : Before Hydro Test

B : Before M/C

C : Before Commissioning